

The Volatility Cube

8.1 Why a single number is never enough

Chapters 6 and 7 priced options given a volatility, as if it were a single known input. In reality, the market does not quote one volatility per underlying rate; it quotes an entire surface, because implied volatility varies systematically with both the option's expiry and its strike. Understanding the shape of this surface, and how the market represents it compactly, is a prerequisite for calibration, because calibration is precisely the exercise of making a model reproduce this surface.

Key result: the three axes of the rates vol surface

A cap/floor or swaption implied volatility is a function of, at minimum:

- **Expiry** T — when the option fixes.
- **Strike** K — relative to the forward, i.e. moneyness.
- **Tenor** (for swaptions only) — the length of the underlying swap once entered.

Cap/floor volatility is therefore genuinely two-dimensional (expiry $\times$ strike); swaption volatility is three-dimensional (expiry $\times$ strike $\times$ tenor), and the swaption surface is accordingly called the **swaption cube** — an expiry/tenor grid of ATM volatilities, each carrying its own strike-dependent smile.

8.2 The smile

At a fixed expiry and tenor, implied volatility plotted against strike is almost never flat. The typical shape — higher implied vol away from the money on both sides, and often asymmetric between low and high strikes — is the **volatility smile** (or **skew**, when the asymmetry dominates). Its existence is itself evidence against the very models this book builds *as stated in their simplest form*: a pure lognormal or pure normal model, with one constant σ or σ_n per underlying, by construction produces a perfectly flat implied volatility across strike. A market smile is the market saying the true distribution of the rate has fatter tails, or more skew, than either simple model allows.

ON THE DESK

Two separate reasons produce a rates smile in practice, and it matters which one dominates for a given currency and regime. First, genuine **skewness** in the distribution of the underlying rate — rates markets have historically shown more downside-vol-expansion during sharp-rally, low-rate regimes than equity-style “crash up” skew would predict, related to the mechanics of central bank policy and the zero/negative-rate boundary discussed in Chapter 1. Second, genuine **stochastic volatility** — volatility itself is random and can spike independently of the direction the rate moves, which fattens both tails symmetrically. A pure displaced-diffusion model (Chapter 15) captures the first effect; a stochastic-volatility extension like SABR captures both.

8.3 SABR: the market's own smile model

The **SABR model** (Stochastic Alpha, Beta, Rho — Hagan, Kumar, Lesniewski and Woodward, 2002) has become, since its publication, the de facto industry-standard tool for *describing* a rates smile, used even by desks whose actual pricing model for exotics is something else entirely (such as BGM). It is worth understanding on its own terms before Chapter 15 discusses how it can be coupled to BGM directly.

Key result: the SABR dynamics

Under the relevant forward measure, SABR models a forward rate F_t (a caplet forward or a swap rate, depending on context) together with a stochastic volatility process α_t :

$$dF_t = \alpha_t F_t^\beta dW_t, \quad d\alpha_t = \nu \alpha_t dZ_t, \quad dW_t dZ_t = \rho dt$$

with four parameters: α_0 (the initial volatility level), $\beta \in [0, 1]$ (the backbone exponent — $\beta = 1$ recovers pure lognormal, $\beta = 0$ recovers pure normal), ρ (the correlation between rate and volatility, which drives skew), and ν (volatility-of-volatility, which drives the symmetric wing of the smile).

Hagan et al. derived a closed-form asymptotic approximation for the Black implied volatility that these dynamics generate as a function of strike, which is why SABR is computationally cheap enough to calibrate to an entire smile in real time and is what almost every trading system uses to *interpolate and interpret* a discretely-quoted market smile, whether or not SABR itself is the pricing engine for anything exotic.

Worked example 8.1 — reading a SABR-calibrated smile.

A $1y \times 5y$ swaption has been fit with SABR parameters $\beta = 0.5$ (fixed by convention, not calibrated), giving a calibrated $\alpha_0 = 0.0092$, $\rho = -0.28$, $\nu = 0.42$, against a forward swap rate of $S = 3.90\%$. A trader wants to know, roughly, whether this smile is skewed toward higher vols for low strikes (payer-hedge-cheap, receiver-hedge-expensive) or the reverse, without running the full Hagan formula.

Step 1 — read ρ directly. The sign of ρ is, to first order, the sign of the skew: $\rho < 0$ means the rate and its volatility are negatively correlated, so volatility tends to rise as the rate falls, which is exactly the pattern of higher implied vol at low strikes — often described as receivers being “bid” relative to payers in vol terms.

Step 2 — read ν for wing thickness. $\nu = 0.42$ is a moderate-to-high vol-of-vol; this widens the smile symmetrically at both wings relative to a low- ν market, on top of the skew ρ already introduces.

Sanity check. A trader who has looked at many SABR fits on this underlying recognises $\rho \approx -0.3$ as a fairly standard, moderate negative skew for a mid-curve USD swaption in a normal-to-elevated rate environment — not an extreme reading in either direction.

Interpretation. Without evaluating the Hagan formula at any specific strike, the trader already knows the qualitative shape: implied vol will be higher for a receiver struck well below 3.90% than for a payer struck the same distance above it. This qualitative reading — sign of ρ for skew direction, size of ν for wing thickness — is the genuinely useful mental shortcut SABR parameters provide, distinct from the closed-form formula itself, which is mechanical once the parameters are fixed.

PITFALL

β is very often *fixed by convention* (commonly $\beta = 0.5$, or $\beta = 0$ to work in pure normal terms) rather than calibrated, because β and ρ trade off against each other in fitting the same smile shape — many (β, ρ) combinations fit an observed smile almost equally well, a classic calibration-degeneracy problem. Treating a fitted β as if it carried independent economic meaning (“the market believes rates are $\beta = 0.7$ lognormal”) over-interprets a number that is frequently pinned by convention rather than genuinely discovered by the optimizer. The same degeneracy problem, in a more consequential form, reappears in Chapter 17 when calibrating BGM’s own volatility and correlation parameters jointly.

8.4 From the cube to the model

The volatility cube is the market’s own description of option prices — it is not, by itself, a description of how the entire forward-rate curve moves jointly through time, which is what a BGM model needs in order to price anything that depends on more than one point on the curve at once (a Bermudan swaption, a spread option, a CMS product). This is the entire reason a term-structure model like BGM is needed at all: to interpolate, extrapolate, and impose joint dynamical consistency on top of a set of prices — caps and swaptions — that the market only quotes one underlying at a time. Parts III and IV build exactly that model; Part V builds the machine that fits it to the cube described in this chapter.

Chapter benchmark

Before moving on, you should be able to, without notes:

1. State the axes of the cap/floor surface and the swaption cube, and explain why swaptions need one more axis than caps.
2. Explain, in words, what a volatility smile is and why a flat one-parameter model cannot produce one.
3. State the SABR dynamics and explain, without doing the algebra, what ρ and ν each control in the resulting smile shape.

